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Engineer to Excel

TECHNICAL QUESTIONS FOR UST GLOBAL – SET1

1. Given a sentence with numbers representing a word's location in the sentence, embedded within each word, and return the sorted sentence.

Note: We are using a maximum of 0-9 numbers only for 1 sentence

Example 1

Input: is1 Thi0s T3est 2a

Output: This is a Test

Example 2

Input: t2o j3oin 4WonderBiz I0 Technolog5ies wan1t

Output: I want to join WonderBiz Technologies

2. Write a program that takes an integer M and M integer array elements as input. The program needs to find the minimum difference between two elements in the integer array. The program then needs to print all those pairs of elements that have the minimum difference. If more than one pair has the minimum difference, then the program should print the output in the ascending order, if an element exists in two or more pairs, then it should be printed two times or more.

Example 1

Input: 4

55 44 33 22

Output: 22 33 33 44 44 55

Explanation: The minimum difference between two elements is 11. Hence the pairs are printed in the ascending order. Here 33 and 44 appear in two different pairs; hence both are printed twice.

Example 2

Input: 5

1 99 22 44 1001

Output: 1 22

3. Write a program to convert a number to binary format.

Example 1

Input: 0

Output: 0

Example 2

Input: 1

Output: 1

4. Write a program that receives a word A and some texts as input. You need to output the texts (without modifying them) in the ascending order of the number of occurrences of the word A in the texts. The input is as follows: an integer M(between 1 and 100, inclusive), followed by the word A in the next line, and some text in each of the M next lines.

Note: The texts and the word A contain only lowercase Latin letters (a,b,c…,z) and blank spaces (“ ”). The maximum size of the texts and the word A is 100 Characters. Every text has a different number of occurrences of the word A.

Note 2:you must print one text per line without modifying the texts.

Example 1

Input: 2

Java

I hate java

Python is a good programming language

Output: Python is a good programming language

I hate java

Example 2

Input: 3

python

I like to code in python

python is named after a show name monty python and not after the snake python

I think python is good i think python is important than php

Output: i like to code in python

i think python is good i think python is important than php

python is named after a show name monty python and not after the snake python

5. Given a string s, Your task is to complete the function encode that returns the run length encoded string for the given string.

eg if the input string is “wwwwaaadexxxxxx”, then the function should return “w4a3d1e1x6″.

You are required to complete the function encode that takes only one argument the string which is to be encoded and returns the encoded string.

Example 1:

Input:

s = aaaabbbccc

Output: a4b3c3

Explanation: a repeated 4 times consecutively b 3 times, c also 3 times.

Example 2:

Input:

s = abbbcdddd

Output: a1b3c1d4

Explanation: a repeated 1 time, b 3 times, c 1 time and d repeated 4 times.

Constraints:

1<=(n = length of s)<=106

6. Given an array arr[] of N distinct numbers, the task is to sort all even-placed numbers in increasing and odd-place numbers in decreasing order. The modified array should contain all sorted even-placed numbers followed by reverse sorted odd-placed numbers.

Note that the first element is considered as even because of its index 0.

Example 1:

Input: N = 8

arr[] = {0, 1, 2, 3, 4, 5, 6, 7}

Output: 0 2 4 6 7 5 3 1

Explanation:

Even-place elements : 0, 2, 4, 6

Odd-place elements : 1, 3, 5, 7

Even-place elements in increasing order :

0, 2, 4, 6

Odd-Place elements in decreasing order :

7, 5, 3, 1

Example 2:

Input: N = 9

arr[] = {3, 1, 2, 4, 5, 9, 13, 14, 12}

Output: 2 3 5 12 13 14 9 4 1

Explanation:

Even-place elements : 3, 2, 5, 13, 12

Odd-place elements : 1, 4, 9, 14

Even-place elements in increasing order :

2, 3, 5, 12, 13

Odd-Place elements in decreasing order :

14, 9, 4, 1

Constraints:

$1 \leq N \leq 105$